

Barite Weight-Up & Rheology Sign-Off Procedure (illustrative)

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1. Purpose & Application

Establishes standardized procedures for synthetic oil-based mud (SOBM) density adjustments using API high-grade barite (4.20 g/cm^3) in deepwater drilling systems. Specifically governs the 11.20 ppg to 11.65 ppg weight-up at 4,172 m MD.

2. Theoretical Formula & Volume Calculations

Mud weight increase formula:

$$\text{Barite (lb/bbl)} = 1470 \times \frac{W_2 - W_1}{35 - W_2}$$

Where:

- $W_1 = 11.20 \text{ ppg}$ (current density)
- $W_2 = 11.65 \text{ ppg}$ (target density)
- Barite requirement $= 28.3 \text{ lb/bbl}$ of active system.

For MN-SM-DW-01 active pit volume of 3,100 bbl:

- Total Barite Required: **87,700 lb** (39.8 metric tons / 796 bags of 50 kg).
- Expected Pit Volume Gain: **60 bbl**.

3. Operational Mixing Execution

1. Addition Rate Limit:

- Barite addition rate via mixing hopper must not exceed 15 MT/hr to prevent sagging and polymer shock.
- Recommended addition duration: 2.5 to 3.0 hours across ~1.5 full circulations.

2. Rheology Control Targets:

- Plastic Viscosity (SPV): target 28 cP.
- Yield Point (Y.P.): maintain target 18 lb/100 ft².
- Gel Strengths ($\frac{s}{m}$): 12 / 22 lb/100 ft².
- Viscosifier / organophilic clay additions to be balanced simultaneously with barite.

4. Verification & Sign-Off Criteria

- Mud balance density readings taken every 15 minutes at suction and flowline.
- Drilling SHALL NOT resume until suction and flowline density show parity at $11.65 \pm 0.05 \text{ ppg}$ for one complete bottoms-up cycle.
- Mud Chemist and Drilling Superintendent sign-off required on MOC Checklist.